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Function Intro
Video: What is a Function? (Algebraically and Graphically) 8 min
Quiz: Functions Quiz Submitted
Domain and Range
Piecewise Functions
What Can You Do to a Function?
Multiplying Binomials
Rationalizing Denominators
Exponent Rules
Logarithms

Functions Quiz

Review Learning Objectives

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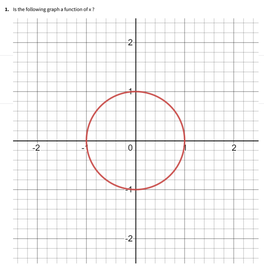
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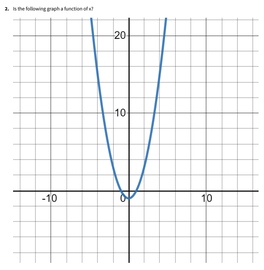
1. Is the following graph a function of x? **1 / 1 point**



- ○ Yes
- ◉ No

Correct
For a curve to be called a graph of a function, it should pass vertical line test. In this case, if a vertical line is drawn on the curve, if the point intersects the curve more than once, then curve does not represent a function.

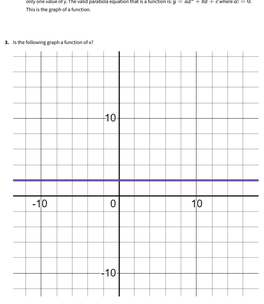
2. Is the following graph a function of x? **1 / 1 point**



- ◉ Yes
- ○ No

Correct
This defines a relation which is a function. For instance, when x is 0, y is unique. Recall y = f(x). A function f(x) needs to have one to one mapping between X (x1, x2...) and Y (y1,y2...). Each point x should have only one point y from f(x). This parabola function passes vertical line test and for each value of x, there exists only one value of y. The valid parabola equation that is a function is $y = ax^2 + bx + c$ where $a^2 = 0$. This is the graph of a function.

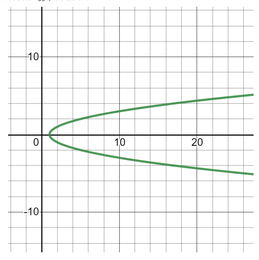
3. Is the following graph a function of x? **1 / 1 point**



- ◉ Yes
- ○ No

Correct
A graph of a function for function f(x) = y such that for each value of x, there is atmost one value of y. It consists of a set of ordered pairs $(x, f(x))$ where x and $f(x)$ are real numbers and are Cartesian coordinates of points in in a two dimensional space. Here for all values of x, y = 2. So this is a graph of a function.

4. Is the following graph a function of x? **1 / 1 point**



- ○ Yes
- ◉ No

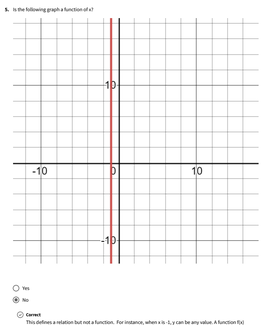
Correct
The equation of a sideways parabola is.

$$x = ay^2 + by + c$$

Notice that it is a quadratic in y.

A function f(x) needs to have one to one mapping between Domain X (x1, x2...) and Codomain Y (y1,y2...). Each point x should have atmost one point y from f(x). And for a curve to be called a graph of a function, it should pass vertical line test. In this case, this parabola function does **"not"** pass vertical line test and for each value of x, there exists **more than one value of y**. The valid parabola equation that is a function that passes vertical line test is: $y = ax^2 + bx + c$ where $a^2 = 0$. We can say U shaped or inverted U shaped parabolas are valid functions and are graphs of respective functions. Sideways parabolas are not graphs of a function.

5. Is the following graph a function of x? **1 / 1 point**



- ○ Yes
- ◉ No

Correct
This defines a relation but not a function. For instance, when x is -1, y can be any value. A function f(x) needs to have one to one mapping between Domain X (x1, x2...) and Codomain Y (y1,y2...). Each point x should have atmost one point y from f(x). In this graph, for each x, there many more values of y that exist. So this is not a graph of the function.

6. Compute $f(0)$ for $f(x) = \frac{x-3}{x+2}$ **1 / 1 point**

$f(0) =$____

Fill in the blanks with your answer. If an answer is undefined then write undefined. Be sure to round off to 3 decimal places if needed. Use the dot to indicate where the decimal goes.

Valid Answer format examples: 1.009, 0.005, 5.224, undefined, 2, 8, 32, 0

Wrong Answer format Examples: 0.0049, 5.2238, .0

-1.5

Correct

$$f(x) = \frac{x-3}{x+2}$$
$$f(0) = \frac{0-3}{0+2} = \frac{-3}{2} = -1.5$$

7. Compute $f(3)$ for $f(x) = \frac{x-3}{x+2}$ **1 / 1 point**

$f(3) =$____

Fill in the blanks with your answer. If an answer is undefined then write undefined. Be sure to round off to 3 decimal places if needed. Use the dot to indicate where the decimal goes.

Valid Answer format examples: 1.009, 0.005, 5.224, undefined, 2, 8, 32, 0

Wrong Answer format Examples: 0.0049, 5.2238, .0

0

Correct

$$f(x) = \frac{x-3}{x+2}$$
$$f(3) = \frac{3-3}{3+2} = \frac{0}{5} = 0$$

8. Compute $f(-2)$ for $f(x) = \frac{x-3}{x+2}$ **1 / 1 point**

$f(-2) =$____

Fill in the blanks with your answer. If an answer is undefined then write undefined. Be sure to round off to 3 decimal places if needed. Use the dot to indicate where the decimal goes.

Valid Answer format examples: 1.009, 0.005, 5.224, undefined, 2, 8, 32, 0

Wrong Answer format Examples: 0.0049, 5.2238, .0

undefined

Correct

$$f(x) = \frac{x-3}{x+2}$$
$$f(3) = \frac{-2-3}{-2+2} = \frac{-6}{0} = undefined$$

Divide by zero is undefined.

9. Compute $f(0)$ rounded off to 3 decimal places for $f(t) = 4e^{-2t}$ **1 / 1 point**

$f(0) =$____

Fill in the blanks with your answer. If an answer is undefined then write undefined. Be sure to round off to 3 decimal places if needed. Use the dot to indicate where the decimal goes.

Valid Answer format examples: 1.009, 0.005, 5.224, undefined, 2, 8, 32, 0

Wrong Answer format Examples: 0.0049, 5.2238, .0

4

Correct

$$f(0) = 4e^{-2t} = 4e^{-2*0} = 4 * e^0 = 4.$$

Since $e^0 = 1$

10. Compute $f(4)$ rounded off to 3 decimal places for $f(t) = 4e^{-2t}$ **1 / 1 point**

$f(4) =$____

Fill in the blanks with your answer. If an answer is undefined then write undefined. Be sure to round off to 3 decimal places if needed. Use the dot to indicate where the decimal goes.

Valid Answer format examples: 1.009, 0.005, 5.224, undefined, 2, 8, 32, 0

Wrong Answer format Examples: 0.0049, 5.2238, .0

0.001

Correct

$$f(4) = 4e^{-2t} = 4e^{-2*4} = 4 * e^{-8} = 0.001.$$

Since $e^{-8} = 0.000335462627$ and $4 * 0.00033 = 0.00132$, rounding off to 3 decimal places we get, .001

11. Compute $f(-2)$ rounded off to 3 decimal places for $f(t) = 4e^{-2t}$ **1 / 1 point**

$f(-2) =$____

Fill in the blanks with your answer. If an answer is undefined then write undefined. Be sure to round off to 3 decimal places if needed. Use the dot to indicate where the decimal goes.

Valid Answer format examples: 1.009, 0.005, 5.224, undefined, 2, 8, 32, 0

Wrong Answer format Examples: 0.0049, 5.2238, .0

218.393

Correct

$$f(-2) = 4e^{-2t} = 4e^{-2*-2} = 4 * e^4 = 218.393.$$

Since $e^8 = 54.598150033$1 and rounding off the actual: $4e^4 = 4 * 54.59815 = 218.3926$ and rounding off to 3 decimal places, we get, 218.393.

12. Compute $f(0)$ for $f(x) = \sqrt{x+6}$ (rounded off to 3 decimal places if needed) **1 / 1 point**

$f(0) =$____

Fill in the blanks with your answer. If an answer is undefined then write undefined. Be sure to round off to 3 decimal places if needed. Use the dot to indicate where the decimal goes.

Valid Answer format examples: 1.009, 0.005, 5.224, undefined, 2, 8, 32, 0

Wrong Answer format Examples: 0.0049, 5.2238, .0

2.449

Correct

$$f(0) = \sqrt{x+6} = \sqrt{0+6} = \sqrt{6} = 2.44948974278$$

Rounding off the value 2.4494 to 3 decimal places, we get 2.449

13. Compute $f(10)$ for $f(x) = \sqrt{x+6}$ (rounded off to 3 decimal places if needed) $f(10) =$____ **1 / 1 point**

Fill in the blanks with your answer. If an answer is undefined then write undefined. Be sure to round off to 3 decimal places if needed. Use the dot to indicate where the decimal goes.

Valid Answer format examples: 1.009, 0.005, 5.224, undefined, 2, 8, 32, 0

Wrong Answer format Examples: 0.0049, 5.2238, .0

4

Correct

$$f(0) = \sqrt{x+6} = \sqrt{10+6} = \sqrt{16} = 4$$

Answer is 4.

14. Compute $f(-10)$ for $f(x) = \sqrt{x+6}$ (rounded off to 3 decimal places if needed) $f(-10) =$____ **1 / 1 point**

Fill in the blanks with your answer. If an answer is undefined then write undefined. Be sure to round off to 3 decimal places if needed. Use the dot to indicate where the decimal goes.

Valid Answer format examples: 1.009, 0.005, 5.224, undefined, 2, 8, 32, 0

Wrong Answer format Examples: 0.0049, 5.2238, .0

undefined

Correct
Correct.

$$f(0) = \sqrt{x+6} = \sqrt{-10+6} = \sqrt{-4} = undefined$$

Since square root of a negative value is undefined. $\sqrt{-1}$ is a Unit Imaginary number belonging to Complex number space. Answer is undefined.